An Elementary Data Defect Audit for Simple OLS Regression

Abstract. Xiao-Li Meng's 2018 paper[1] on the Big Data Paradox introduced the data defect correlation as a finite-population measure of the degree to which inclusion in a dataset is associated with the quantity being estimated. In the simplest case, where a recorded sample mean is used to estimate a population mean, Meng showed that the resulting error factors exactly into a data quality term, a data quantity term, and a problem difficulty term. This note asks whether the same idea can be carried into ordinary least squares regression in a straightforward and useful way.

Perhaps the most simpleminded approach is to observe that the simple OLS slope is itself a smooth function of several finite population means, so that Meng's data defect correlation may be applied first to the primitive moment functions entering the coefficient, namely X, Y, X^2, and XY, after which the resulting marginal moment defects may be propagated through the OLS functional by a first-order Taylor approximation via an approach analogous to the delta method. This perturbation-propagation method yields a finite-population diagnostic for how construction of an analytic regression sample changes the slope relative to a benchmark frame (ideally, but not necessarily, a census).

1. Review of Meng's identity

Let $\mathcal{P} = \{1, \ldots, N\}$ denote a finite population, and suppose that the quantity of interest may be written as the population mean of a scalar function G, i.e.,

$$\overline{G}_N = \frac{1}{N} \sum_{k=1}^{N} G_k.$$

Let $\mathcal{A} \subseteq \mathcal{P}$ denote the actually recorded, practically accessible, or analytically retained units, and define the recording indicator

$$R_k := \mathbf{1}[k \in \mathcal{A}], \quad n = \sum_{k=1}^{N} R_k, \quad f = \frac{n}{N}.$$

The corresponding observed data average is then just the usual sample mean

$$\overline{G}_n = \frac{\sum_{k=1}^{N} R_k G_k}{\sum_{k=1}^{N} R_k} = \frac{1}{n} \sum_{j \in \mathcal{A}} G_j.$$

Following Meng, we introduce a uniformly distributed random index $J \sim \mathcal{U}\{1, \ldots, N\}$, which permits us to rewrite finite population averages as expectations with respect to a uniformly selected population label as

$$\overline{G}_N = \mathrm{E}_J[G_J] \quad \text{and} \quad \overline{G}_n = \mathrm{E}[G_J \mid R_J = 1] = \frac{\mathrm{E}_J[R_J G_J]}{\mathrm{E}_J[R_J]},$$

so that

$$\overline{G}_n - \overline{G}_N = \frac{\mathrm{E}_J[R_J G_J]}{\mathrm{E}_J[R_J]} - \mathrm{E}_J[G_J] = \frac{\mathrm{E}_J[R_J G_J] - \mathrm{E}_J[R_J]\mathrm{E}_J[G_J]}{\mathrm{E}_J[R_J]} = \frac{\mathrm{Cov}_J(R_J, G_J)}{\mathrm{E}_J[R_J]}.$$

Since $R_j \sim \text{Bernoulli}(f)$ for any fixed R-mechanism, we have $\mathrm{E}_J[R_J] = f$ and $\mathrm{Var}_J(R_J) = f(1-f)$. Writing

$$\rho_{R,G} := \mathrm{Corr}_J(R_J, G_J) \quad \text{and} \quad \sigma_G^2 := \mathrm{Var}_J(G_J) = \frac{1}{N}\sum_{k=1}^{N}(G_k - \overline{G}_N)^2,$$

we obtain Meng's data defect identity

$$\overline{G}_n - \overline{G}_N = \frac{\mathrm{Cov}_J(R_J, G_J)}{f\sqrt{f(1-f)}\sigma_G}\sqrt{f(1-f)}\,\sigma_G = \underbrace{\rho_{R,G}}_{\text{data quality}} \underbrace{\sqrt{\frac{1-f}{f}}}_{\text{data quantity}} \underbrace{\sigma_G}_{\text{problem difficulty}}.$$

Thus it is seen that the sample quality metric of interest is not the sample size alone. The quantity $\rho_{R,G}$ measures the extent to which inclusion in the sample data is associated with the variable whose finite-population mean is being estimated. A very large dataset may therefore be misleading if R is even weakly correlated with G, particularly when the target population is large and G is variable.

To demonstrate how badly things can go wrong in the large N regime, we consider Meng's main scaling argument obtained by comparing the actual error above to the benchmark error one would expect under simple random sampling. The usual finite-population variance with denominator $N-1$ is

$$S_{G,\mathrm{SRS}}^2 := \frac{1}{N-1}\sum_{k=1}^{N}(G_k - \overline{G}_N)^2 = \frac{N}{N-1}\sigma_G^2.$$

Under simple random sampling without replacement of $n = fN$ units from $\mathcal{P}$, the benchmark variance of the sample mean is therefore

$$\mathrm{Var}_{\mathrm{SRS}}(\bar{G}_n) = \frac{1-f}{n}S_{G,\mathrm{SRS}}^2 = \frac{1-f}{fN}\cdot\frac{N}{N-1}\sigma_G^2 = \frac{1-f}{f(N-1)}\sigma_G^2,$$

so that the nominal score obtained by measuring the actual error against the simple-random-sampling benchmark is given by

$$Z_{n,N} := \frac{\overline{G}_n - \overline{G}_N}{\sqrt{\mathrm{Var}_{\mathrm{SRS}}(\bar{G}_n)}} = \frac{\rho_{R,G}\sqrt{\frac{1-f}{f}}\sigma_G}{\sqrt{\frac{1-f}{f(N-1)}}\sigma_G} = \sqrt{N-1}\rho_{R,G}.$$

Thus, to keep the realized error comparable to ordinary sampling error, the data defect correlation must be of order $N^{-1/2}$, quite a strong condition for the large N observed in Big Data contexts.

The foregoing development provides a quantitatively precise formulation of how probability sampling controls data quality: Under a well-designed sampling mechanism, $\rho_{R,G}$ is forced to fluctuate at roughly the $N^{-1/2}$ scale. A non-random administrative or convenience sample, by contrast, *need not exhibit this cancellation.* Consequently, even a numerically tiny data defect correlation may correspond to a large nominal error when N is large.

2. Applying the identity to the moments underlying simple OLS

Now suppose that each population unit has an outcome Y_k and a scalar predictor X_k. The finite-population simple OLS slope of Y on X is

$$\beta_{\mathcal{P}} = \frac{\mathrm{Cov}_{\mathcal{P}}(X, Y)}{\mathrm{Var}_{\mathcal{P}}(X)}.$$

Writing the relevant finite population moments as

$$\mu_x := \mathrm{E}_{\mathcal{P}}[X], \quad \mu_y = \mathrm{E}_{\mathcal{P}}[Y], \quad \mu_{x^2} = \mathrm{E}_{\mathcal{P}}[X^2], \quad \mu_{xy} = \mathrm{E}_{\mathcal{P}}[XY],$$

we may express the population slope as

$$\beta_{\mathcal{P}} = \frac{\mu_{xy} - \mu_x \mu_y}{\mu_{x^2} - \mu_x^2}.$$

The sample slope β_n is obtained by replacing each of these finite population moments with its corresponding analytic sample moment. Hence the primitive defects are

$$\delta_x = \overline{X}_n - \overline{X}_N, \quad \delta_y = \overline{Y}_n - \overline{Y}_N, \quad \delta_{x^2} = \overline{X^2}_n - \overline{X^2}_N, \quad \delta_{xy} = \overline{XY}_n - \overline{XY}_N.$$

By Meng's identity, each of these defects has an exact DDC representation in terms of the moment perturbations

$$\delta_x = \rho_{R,X} \sqrt{\frac{1-f}{f}}\, \sigma_X,$$

$$\delta_y = \rho_{R,Y} \sqrt{\frac{1-f}{f}}\, \sigma_Y,$$

$$\delta_{x^2} = \rho_{R,X^2} \sqrt{\frac{1-f}{f}}\, \sigma_{X^2},$$

$$\delta_{xy} = \rho_{R,XY} \sqrt{\frac{1-f}{f}} \sigma_{XY}.$$

The proposed diagnostic begins by asking whether inclusion in the the sample $\mathcal{A}$ is correlated with the four scalar moment functions X, Y, X^2, and XY. These are the quantities that determine the simple OLS slope.

3. First-order propagation to the OLS coefficient

Define

$$C_{\mathcal{P}} := \mu_{xy} - \mu_x \mu_y, \quad V_{\mathcal{P}} := \mu_{x^2} - \mu_x^2,$$

so that

$$\beta_{\mathcal{P}} = \frac{C_{\mathcal{P}}}{V_{\mathcal{P}}}.$$

Passing from the benchmark population $\mathcal{P}$ to the analytic sample $\mathcal{A}$ replaces the primitive moments by

$$\mu_x + \delta_x, \quad \mu_y + \delta_y, \quad \mu_{x^2} + \delta_{x^2}, \quad \mu_{xy} + \delta_{xy}$$

so that the sample covariance numerator is

$$\begin{aligned} C_{\mathcal{A}} &= (\mu_{xy} + \delta_{xy}) - (\mu_x + \delta_x)(\mu_y + \delta_y) \\ &= \mu_{xy} + \delta_{xy} - \mu_x\mu_y - \mu_y\delta_x - \mu_x\delta_y - \delta_x\delta_y \\ &= C_{\mathcal{P}} + \delta_{xy} - \mu_y\delta_x - \mu_x\delta_y - \delta_x\delta_y. \end{aligned}$$

Thus

$$\delta_C := C_{\mathcal{A}} - C_{\mathcal{P}} = \delta_{xy} - \mu_y\delta_x - \mu_x\delta_y - \delta_x\delta_y.$$

Discarding the second-order product $\delta_x\delta_y$, we obtain the first-order approximation

$$\delta_C \approx \delta_{xy} - \mu_y\delta_x - \mu_x\delta_y.$$

Similarly, the sample variance denominator is

$$\begin{aligned} V_{\mathcal{A}} &= (\mu_{x^2} + \delta_{x^2}) - (\mu_x + \delta_x)^2 \\ &= \mu_{x^2} + \delta_{x^2} - \mu_x^2 - 2\mu_x\delta_x - \delta_x^2 \\ &= V_{\mathcal{P}} + \delta_{x^2} - 2\mu_x\delta_x - \delta_x^2. \end{aligned}$$

Thus

$$\delta_V := V_{\mathcal{A}} - V_{\mathcal{P}} = \delta_{x^2} - 2\mu_x\delta_x - \delta_x^2 \approx \delta_{x^2} - 2\mu_x\delta_x,$$

where we have again taken a first-order approximation in the perturbations. The sample slope may therefore be written as

$$\beta_{\mathcal{A}} = \frac{C_{\mathcal{A}}}{V_{\mathcal{A}}} = \frac{C_{\mathcal{P}} + \delta_C}{V_{\mathcal{P}} + \delta_V}.$$

Subtracting $\beta_{\mathcal{P}} = C_{\mathcal{P}}/V_{\mathcal{P}}$ then gives the exact identity

$$\begin{aligned} \beta_{\mathcal{A}} - \beta_{\mathcal{P}} &= \frac{C_{\mathcal{P}} + \delta_C}{V_{\mathcal{P}} + \delta_V} - \frac{C_{\mathcal{P}}}{V_{\mathcal{P}}} \\ &= \frac{V_{\mathcal{P}}(C_{\mathcal{P}} + \delta_C) - C_{\mathcal{P}}(V_{\mathcal{P}} + \delta_V)}{V_{\mathcal{P}}(V_{\mathcal{P}} + \delta_V)} \\ &= \frac{V_{\mathcal{P}}\delta_C - C_{\mathcal{P}}\delta_V}{V_{\mathcal{P}}(V_{\mathcal{P}} + \delta_V)}. \end{aligned}$$

Since $C_{\mathcal{P}} = \beta_{\mathcal{P}}V_{\mathcal{P}}$, this becomes

$$\beta_{\mathcal{A}} - \beta_{\mathcal{P}} = \frac{\delta_C - \beta_{\mathcal{P}}\delta_V}{V_{\mathcal{P}} + \delta_V} = \frac{\delta_C - \beta_{\mathcal{P}}\delta_V}{V_{\mathcal{P}}} - \frac{(\delta_C - \beta_{\mathcal{P}}\delta_V)\delta_V}{V_{\mathcal{P}}^2} + \mathcal{O}(\delta^3) = \frac{\delta_C - \beta_{\mathcal{P}}\delta_V}{V_{\mathcal{P}}} + \mathcal{O}(\delta^2).$$

When the moment defects are small relative to the benchmark denominator, we replace $V_{\mathcal{P}} + \delta_V$ by $V_{\mathcal{P}}$ (see section **7** below for a more rigorous derivation using Taylor's theorem), yielding the first-order ratio approximation

$$\beta_{\mathcal{A}} - \beta_{\mathcal{P}} \approx \frac{\delta_C - \beta_{\mathcal{P}}\delta_V}{V_{\mathcal{P}}}.$$

Substituting the first-order expressions for δ_C and δ_V gives

$$\begin{aligned}\beta_{\mathcal{A}} - \beta_{\mathcal{P}} &\approx \frac{(\delta_{xy} - \mu_y\delta_x - \mu_x\delta_y) - \beta_{\mathcal{P}}(\delta_{x^2} - 2\mu_x\delta_x)}{V_{\mathcal{P}}} \\ &= \frac{\delta_{xy} - \mu_y\delta_x - \mu_x\delta_y - \beta_{\mathcal{P}}\delta_{x^2} + 2\beta_{\mathcal{P}}\mu_x\delta_x}{V_{\mathcal{P}}} \\ &= \frac{\delta_{xy} + (2\beta_{\mathcal{P}}\mu_x - \mu_y)\delta_x - \mu_x\delta_y - \beta_{\mathcal{P}}\delta_{x^2}}{V_{\mathcal{P}}}.\end{aligned}$$

This is the basic proposed extension. Regression coefficient distortion is represented as a first-order consequence of defects in the primitive moment functions. Since each moment defect is determined by a corresponding data defect correlation, a data quantity term, and a problem difficulty term, the regression coefficient inherits its distortion from the 'marginal' DDCs associated with X, Y, X^2, and XY.

4. A coefficient-level defect score

The preceding expression also suggests a useful coefficient-level diagnostic. Although $\beta_{\mathcal{P}}$ itself is not a scalar finite-population attribute of the individual units, its first-order perturbation may be represented as the mean defect of a scalar unit-level quantity.

Let

$$V_X := \mu_{x^2} - \mu_x^2,$$

and define the benchmark-centered coefficient score

$$\psi_k := \frac{(X_k - \mu_x)(Y_k - \mu_y) - \beta_{\mathcal{P}}(X_k - \mu_x)^2}{V_X}.$$

Since $\beta_{\mathcal{P}} = \mathrm{Cov}_{\mathcal{P}}(X, Y)/\mathrm{Var}_{\mathcal{P}}(X)$, we have

$$\begin{aligned}\mathrm{E}_{\mathcal{P}}[\psi_k] &= \frac{\mathrm{E}_{\mathcal{P}}[(X_k - \mu_x)(Y_k - \mu_y)] - \beta_{\mathcal{P}}\mathrm{E}_{\mathcal{P}}[(X_k - \mu_x)^2]}{V_X} \\ &= \frac{\mathrm{Cov}_{\mathcal{P}}(X, Y) - \beta_{\mathcal{P}}V_X}{V_X} = 0.\end{aligned}$$

Now compute the sample mean defect in ψ. Since μ_x, μ_y, $\beta_{\mathcal{P}}$, and V_X are held fixed at their benchmark values, we have

$$\bar{\psi}_{\mathcal{A}} - \bar{\psi}_{\mathcal{P}} = \bar{\psi}_{\mathcal{A}} = \frac{\mathrm{E}_{\mathcal{A}}[(X_k - \mu_x)(Y_k - \mu_y)] - \beta_{\mathcal{P}}\mathrm{E}_{\mathcal{A}}[(X_k - \mu_x)^2]}{V_X}.$$

The first term in the numerator satisfies

$$\begin{aligned}\mathrm{E}_{\mathcal{A}}[(X_k-\mu_x)(Y_k-\mu_y)] &= \mathrm{E}_{\mathcal{A}}[X_kY_k]-\mu_x\mathrm{E}_{\mathcal{A}}[Y_k]-\mu_y\mathrm{E}_{\mathcal{A}}[X_k]+\mu_x\mu_y\\ &= (\mu_{xy}+\delta_{xy})-\mu_x(\mu_y+\delta_y)-\mu_y(\mu_x+\delta_x)+\mu_x\mu_y\\ &= \mathrm{Cov}_{\mathcal{P}}(X,Y)+\delta_{xy}-\mu_x\delta_y-\mu_y\delta_x.\end{aligned}$$

Similarly,

$$\begin{aligned}\mathrm{E}_{\mathcal{A}}[(X_k-\mu_x)^2] &= \mathrm{E}_{\mathcal{A}}[X_k^2]-2\mu_x\mathrm{E}_{\mathcal{A}}[X_k]+\mu_x^2\\ &= (\mu_{x^2}+\delta_{x^2})-2\mu_x(\mu_x+\delta_x)+\mu_x^2\\ &= V_X+\delta_{x^2}-2\mu_x\delta_x.\end{aligned}$$

Substituting these expressions gives

$$\begin{aligned}\bar{\psi}_{\mathcal{A}}-\bar{\psi}_{\mathcal{P}} &= \frac{\mathrm{Cov}_{\mathcal{P}}(X,Y)+\delta_{xy}-\mu_x\delta_y-\mu_y\delta_x-\beta_{\mathcal{P}}(V_X+\delta_{x^2}-2\mu_x\delta_x)}{V_X}\\ &= \frac{\delta_{xy}+(2\beta_{\mathcal{P}}\mu_x-\mu_y)\delta_x-\mu_x\delta_y-\beta_{\mathcal{P}}\delta_{x^2}}{V_X},\end{aligned}$$

so that

$$\beta_{\mathcal{A}}-\beta_{\mathcal{P}}\approx\bar{\psi}_{\mathcal{A}}-\bar{\psi}_{\mathcal{P}}.$$

Applying Meng's identity directly to ψ yields

$$\bar{\psi}_{\mathcal{A}}-\bar{\psi}_{\mathcal{P}}=\rho_{R,\psi}\sqrt{\frac{1-f}{f}}\,\sigma_\psi,$$

where

$$\rho_{R,\psi}:=\mathrm{Corr}_{\mathcal{P}}(R_k,\psi_k)\quad\text{and}\quad\sigma_\psi^2:=\mathrm{Var}_{\mathcal{P}}(\psi_k),$$

so that the linearized coefficient defect may be standardized against the simple random sampling benchmark as

$$Z_\beta^{\mathrm{lin}}:=\frac{\bar{\psi}_{\mathcal{A}}-\bar{\psi}_{\mathcal{P}}}{\sqrt{\dfrac{1-f}{f(N-1)}}\sigma_\psi}=\sqrt{N-1}\rho_{R,\psi}.$$

By comparison, the actual standardized coefficient gap is

$$Z_\beta^{\mathrm{act}}:=\frac{\beta_{\mathcal{A}}-\beta_{\mathcal{P}}}{\sqrt{\dfrac{1-f}{f(N-1)}}\sigma_\psi}.$$

The difference between these two quantities could serve as a diagnostic for the adequacy of the first-order approximation. If Z_β^{act} and Z_β^{lin} are close, then the coefficient shift is well described by

the defect in ψ. If they differ substantially, then the analytic sample is no longer behaving like a local perturbation of the benchmark frame, so that higher-order terms, leverage, or substantial changes in the geometry of X are materially affecting the fitted slope.

5. Calibration weighting as a repair diagnostic

The preceding audit indicates which primitive moment defects contribute to the observed slope distortion. A natural follow-up question is whether the analytic sample can be reweighted so as to repair these defects. This leads to a calibration-weighting exercise in the sense of Deville and Särndal's calibration estimators, generalized raking, and related entropy-balancing methods[2]. The purpose here is not to introduce a new weighting estimator, but to use a standard calibration idea as a diagnostic companion to the DDC decomposition.

Let h_k denote a vector of calibration functions observed on the benchmark frame $\mathcal{P}$ and on the analytic sample $\mathcal{A}$. In the present application, possible choices include $h_k = X_k,$, $h_k = (X_k, X_k^2)$, $h_k = (X_k, Y_k)$, or the full primitive moment vector $h_k = (X_k, Y_k, X_k^2, X_kY_k)$. The benchmark target is

$$\bar{h}_{\mathcal{P}} := \frac{1}{N} \sum_{k \in \mathcal{P}} h_k.$$

We then seek weights w_k on the analytic sample satisfying

$$w_k > 0, \quad \sum_{k \in \mathcal{A}} w_k = 1, \quad \sum_{k \in \mathcal{A}} w_k h_k = \bar{h}_{\mathcal{P}}.$$

Among all weights satisfying these constraints, entropy calibration chooses the weights closest to uniform sample weights in Kullback-Leibler divergence. Letting $q_k = 1/n_{\mathcal{A}}$, the optimization problem is

$$\min_{\{w_k\}_{k \in \mathcal{A}}} \sum_{k \in \mathcal{A}} w_k \ln\left(\frac{w_k}{q_k}\right)$$

subject to

$$\sum_{k \in \mathcal{A}} w_k = 1 \quad \text{and} \quad \sum_{k \in \mathcal{A}} w_k h_k = \bar{h}_{\mathcal{P}}.$$

The resulting weighted simple-regression coefficient is computed from the weighted moments

$$\mu_{x,w} = \sum_{k \in \mathcal{A}} w_k X_k, \quad \mu_{y,w} = \sum_{k \in \mathcal{A}} w_k Y_k, \quad \mu_{x^2,w} = \sum_{k \in \mathcal{A}} w_k X_k^2, \quad \mu_{xy,w} = \sum_{k \in \mathcal{A}} w_k X_k Y_k,$$

so that

$$\beta_w = \frac{\mu_{xy,w} - \mu_{x,w}\mu_{y,w}}{\mu_{x^2,w} - \mu_{x,w}^2}.$$

This calibration exercise has a useful interpretation. If the DDC audit suggests that the coefficient defect is driven mainly by δ_x, then calibration on X should move β_w toward $\beta_{\mathcal{P}}$. If the defect is driven mainly by δ_{x^2}, then calibration on X alone may be insufficient, since the problem concerns the spread or tail behavior of X, not only its mean. If the defect is driven mainly by

δ_{xy}, then marginal calibration on X and Y may again be insufficient, since the defect lies in the joint structure that identifies the slope, i.e., deviation in the R-mechanism from simple random sampling is systematically distorting the population association between X and Y.

In the limiting case where the calibration vector is

$$h_k = (X_k, Y_k, X_k^2, X_k Y_k),$$

successful exact calibration forces

$$\mu_{x,w} = \mu_x, \quad \mu_{y,w} = \mu_y, \quad \mu_{x^2,w} = \mu_{x^2}, \quad \mu_{xy,w} = \mu_{xy},$$

and therefore forces

$$\beta_w = \beta_{\mathcal{P}}.$$

The informative question is how costly is this repair in terms of weight instability. To assess this cost, we report the effective sample size

$$n_{\text{eff}}(w) = \frac{1}{\sum_{k \in \mathcal{A}} w_k^2},$$

along with the minimum weight, maximum weight, weight ratio, and maximum absolute balance error. Severe weight concentration indicates that the analytic sample can only be made benchmark-like by placing very large weight on a small number of units, suggesting these units for additional scrutiny in the form of, e.g., leverage/influence analysis, etc.

The FEC calibration results are consistent with this interpretation. In the mild receipt threshold regime `ttl_receipts` $\geq$ 10,000, full primitive moment calibration restores the benchmark slope with little loss of effective sample size and only modest weight dispersion. By contrast, in the more severe regime `ttl_receipts` $\geq$ 100,000, full primitive moment calibration again restores the benchmark slope, but only by reducing the effective sample size from more than six thousand observations to roughly a few hundred and by producing an extremely large maximum-to-minimum weight ratio, indicating that the high-receipts subset has lost so much of the benchmark geometry that the missing structure can be reproduced only by highly concentrated weights.

The same distinction appears in the political subsets. Calibration on X alone only mildly repairs the ruling party slope gap, while full primitive moment calibration restores the benchmark slope at the cost of noticeably increased weight dispersion. In the open-seat subsets, the unweighted slope is already close to the benchmark in the broader open-seat case, while the ruling party open-seat subset requires stronger reweighting to recover the benchmark relationship.

This simple calibration exercise clarifies the practical use of this approximation: The audit identifies the primitive moments responsible for coefficient distortion; the calibration then determines whether those defects are correctable by reweighting the observed analytic sample. If a small set of constraints repairs the slope with stable weights, the sample defect is plausibly a tractable calibration problem. If repair requires extreme weights or a large collapse in effective sample size, the defect is closer to a support problem in that the analytic sample lacks enough of the benchmark frame to represent it without heavy extrapolation via outsized weights.

6. Natural next steps

The same idea extends directly to multiple regression, though the notation becomes more compact in matrix form. Let $\mathbf{x}_k$ denote the column vector of regressors for population unit k. If the regression includes an intercept, then

$$\mathbf{x}_k = \begin{pmatrix} 1 \\ X_{1k} \\ \vdots \\ X_{pk} \end{pmatrix}.$$

The finite-population least squares coefficient is defined by

$$\boldsymbol{\beta}_{\mathcal{P}} := \arg\min_{\boldsymbol{\beta}} \mathrm{E}_{\mathcal{P}}\left[(Y_k - \mathbf{x}_k^\top \boldsymbol{\beta})^2\right].$$

The associated normal equations are obtained by differentiating the population least squares criterion with respect to $\boldsymbol{\beta}$, giving

$$\mathrm{E}_{\mathcal{P}}\left[\mathbf{x}_k(Y_k - \mathbf{x}_k^\top \boldsymbol{\beta}_{\mathcal{P}})\right] = \mathbf{0} \Longleftrightarrow \mathrm{E}_{\mathcal{P}}[\mathbf{x}_k Y_k] = \mathrm{E}_{\mathcal{P}}[\mathbf{x}_k \mathbf{x}_k^\top]\boldsymbol{\beta}_{\mathcal{P}}.$$

Writing

$$\mathbf{Q}_{\mathcal{P}} := \mathrm{E}_{\mathcal{P}}[\mathbf{x}_k \mathbf{x}_k^\top] \quad \text{and} \quad \mathbf{q}_{\mathcal{P}} := \mathrm{E}_{\mathcal{P}}[\mathbf{x}_k Y_k],$$

we then have

$$\boldsymbol{\beta}_{\mathcal{P}} = \mathbf{Q}_{\mathcal{P}}^{-1}\mathbf{q}_{\mathcal{P}},$$

provided $\mathbf{Q}_{\mathcal{P}}$ is nonsingular, so that $\mathbf{Q}_{\mathcal{P}}$ is the finite population second moment matrix of the regressors, while $\mathbf{q}_{\mathcal{P}}$ is the finite population cross-moment vector between the regressors and the outcome.

For example, in the simple regression case with intercept, we have

$$\mathbf{x}_k = \begin{pmatrix} 1 \\ X_k \end{pmatrix}, \quad \mathbf{Q}_{\mathcal{P}} = \begin{pmatrix} 1 & \mu_x \\ \mu_x & \mu_{x^2} \end{pmatrix}, \quad \mathbf{q}_{\mathcal{P}} = \begin{pmatrix} \mu_y \\ \mu_{xy} \end{pmatrix}.$$

Solving

$$\begin{pmatrix} 1 & \mu_x \\ \mu_x & \mu_{x^2} \end{pmatrix} \begin{pmatrix} \alpha_{\mathcal{P}} \\ \beta_{\mathcal{P}} \end{pmatrix} = \begin{pmatrix} \mu_y \\ \mu_{xy} \end{pmatrix}$$

recovers the familiar slope

$$\beta_{\mathcal{P}} = \frac{\mu_{xy} - \mu_x \mu_y}{\mu_{x^2} - \mu_x^2}.$$

Now suppose that passing from the benchmark population $\mathcal{P}$ to the analytic sample $\mathcal{A}$ perturbs the moment matrix and cross-moment vector by

$$\mathbf{Q}_{\mathcal{A}} = \mathbf{Q}_{\mathcal{P}} + \mathrm{d}\mathbf{Q} \quad \text{and} \quad \mathbf{q}_{\mathcal{A}} = \mathbf{q}_{\mathcal{P}} + \mathrm{d}\mathbf{q}.$$

Since

$$\boldsymbol{\beta} = \mathbf{Q}^{-1}\mathbf{q},$$

the first-order differential is

$$\begin{aligned} \mathrm{d}\boldsymbol{\beta} &= \mathrm{d}(\mathbf{Q}^{-1}\mathbf{q}) \\ &= \mathrm{d}(\mathbf{Q}^{-1})\mathbf{q} + \mathbf{Q}^{-1}\mathrm{d}\mathbf{q}. \end{aligned}$$

Using the standard matrix differential identity

$$\mathrm{d}(\mathbf{Q}^{-1}) = -\mathbf{Q}^{-1}(\mathrm{d}\mathbf{Q})\mathbf{Q}^{-1},$$

and evaluating at the benchmark population moments gives

$$\begin{aligned} \mathrm{d}\boldsymbol{\beta} &\approx -\mathbf{Q}_{\mathcal{P}}^{-1}(\mathrm{d}\mathbf{Q})\mathbf{Q}_{\mathcal{P}}^{-1}\mathbf{q}_{\mathcal{P}} + \mathbf{Q}_{\mathcal{P}}^{-1}\mathrm{d}\mathbf{q} \\ &= -\mathbf{Q}_{\mathcal{P}}^{-1}(\mathrm{d}\mathbf{Q})\boldsymbol{\beta}_{\mathcal{P}} + \mathbf{Q}_{\mathcal{P}}^{-1}\mathrm{d}\mathbf{q} \\ &= \mathbf{Q}_{\mathcal{P}}^{-1}(\mathrm{d}\mathbf{q} - (\mathrm{d}\mathbf{Q})\boldsymbol{\beta}_{\mathcal{P}}). \end{aligned}$$

Therefore,

$$\boldsymbol{\beta}_{\mathcal{A}} - \boldsymbol{\beta}_{\mathcal{P}} \approx \mathbf{Q}_{\mathcal{P}}^{-1}((\mathbf{q}_{\mathcal{A}} - \mathbf{q}_{\mathcal{P}}) - (\mathbf{Q}_{\mathcal{A}} - \mathbf{Q}_{\mathcal{P}})\boldsymbol{\beta}_{\mathcal{P}}).$$

To see that this matrix formula reproduces the preceding simple regression calculation, note that in the one-predictor case with intercept we have

$$\mathrm{d}\mathbf{Q} = \begin{pmatrix} 0 & \delta_x \\ \delta_x & \delta_{x^2} \end{pmatrix}, \quad \mathrm{d}\mathbf{q} = \begin{pmatrix} \delta_y \\ \delta_{xy} \end{pmatrix}.$$

The upper-left entry of $\mathrm{d}\mathbf{Q}$ is zero because the intercept regressor is identically equal to 1, and hence has no mean defect. Since

$$\boldsymbol{\beta}_{\mathcal{P}} = \begin{pmatrix} \alpha_{\mathcal{P}} \\ \beta_{\mathcal{P}} \end{pmatrix},$$

we obtain

$$\mathrm{d}\mathbf{q} - (\mathrm{d}\mathbf{Q})\boldsymbol{\beta}_{\mathcal{P}} = \begin{pmatrix} \delta_y \\ \delta_{xy} \end{pmatrix} - \begin{pmatrix} 0 & \delta_x \\ \delta_x & \delta_{x^2} \end{pmatrix}\begin{pmatrix} \alpha_{\mathcal{P}} \\ \beta_{\mathcal{P}} \end{pmatrix} = \begin{pmatrix} \delta_y - \beta_{\mathcal{P}}\delta_x \\ \delta_{xy} - \alpha_{\mathcal{P}}\delta_x - \beta_{\mathcal{P}}\delta_{x^2} \end{pmatrix}.$$

Moreover,

$$\mathbf{Q}_{\mathcal{P}}^{-1} = \frac{1}{\mu_{x^2} - \mu_x^2}\begin{pmatrix} \mu_{x^2} & -\mu_x \\ -\mu_x & 1 \end{pmatrix}$$

so that the second component of $\mathrm{d}\boldsymbol{\beta}$ is

$$\begin{aligned} \mathrm{d}\beta &\approx \frac{-\mu_x(\delta_y - \beta_{\mathcal{P}}\delta_x) + \delta_{xy} - \alpha_{\mathcal{P}}\delta_x - \beta_{\mathcal{P}}\delta_{x^2}}{\mu_{x^2} - \mu_x^2} \\ &= \frac{\delta_{xy} - \mu_x\delta_y + (\mu_x\beta_{\mathcal{P}} - \alpha_{\mathcal{P}})\delta_x - \beta_{\mathcal{P}}\delta_{x^2}}{\mu_{x^2} - \mu_x^2}. \end{aligned}$$

Finally, since

$$\alpha_{\mathcal{P}} = \mu_y - \beta_{\mathcal{P}}\mu_x,$$

we have

$$\mu_x\beta_{\mathcal{P}} - \alpha_{\mathcal{P}} = 2\beta_{\mathcal{P}}\mu_x - \mu_y,$$

so that

$$\mathrm{d}\beta \approx \frac{\delta_{xy} + (2\beta_{\mathcal{P}}\mu_x - \mu_y)\delta_x - \mu_x\delta_y - \beta_{\mathcal{P}}\delta_{x^2}}{\mu_{x^2}\mu_x^2},$$

which is exactly the first-order expression obtained above by directly perturbing the scalar slope formula.

This is the multiple regression analogue of the simple OLS formula above. Each scalar entry of $\mathbf{Q}_{\mathcal{A}} - \mathbf{Q}_{\mathcal{P}}$ is a data defect in a regressor cross-product, while each scalar entry of $\mathbf{q}_{\mathcal{A}} - \mathbf{q}_{\mathcal{P}}$ is a data defect in a regressor outcome cross-product. Thus one may compute Meng-style data defect correlations for the scalar moment functions X_aX_b and X_aY then propagate those defects through the regression functional by the preceding first-order expression.

7. Appendix: Approximation

One may rigorously obtain the first-order coefficient perturbation derived heuristically in **3** above by regarding the slope as a smooth function of the primitive moments. Define

$$\phi(\mu_x, \mu_y, \mu_{x^2}, \mu_{xy}) := \frac{\mu_{xy} - \mu_x\mu_y}{\mu_{x^2} - \mu_x^2}.$$

This map is smooth on the open set where $\mu_{x^2} - \mu_x^2 > 0$. Thus, provided the benchmark variance $V_{\mathcal{P}}$ is positive and the analytic sample perturbation does not move the denominator through zero, the ordinary first-order Taylor expansion gives

$$\beta_{\mathcal{A}} - \beta_{\mathcal{P}} = \phi(\boldsymbol{\mu}_{\mathcal{P}} + \boldsymbol{\delta}) - \phi(\boldsymbol{\mu}_{\mathcal{P}}) = D\phi(\boldsymbol{\mu}_{\mathcal{P}})[\boldsymbol{\delta}] + O(\|\boldsymbol{\delta}\|^2),$$

where

$$\boldsymbol{\mu}_{\mathcal{P}} := (\mu_x, \mu_y, \mu_{x^2}, \mu_{xy}) \quad \text{and} \quad \boldsymbol{\delta} := (\delta_x, \delta_y, \delta_{x^2}, \delta_{xy}).$$

Equivalently, introduce the path

$$\boldsymbol{\mu}(t) = \boldsymbol{\mu}_{\mathcal{P}} + t\boldsymbol{\delta}, \quad 0 \le t \le 1,$$

and write

$$\beta(t) = \phi(\boldsymbol{\mu}(t)) = \frac{C(t)}{V(t)},$$

where

$$C(t) = (\mu_{xy} + t\delta_{xy}) - (\mu_x + t\delta_x)(\mu_y + t\delta_y)$$

and

$$V(t) = (\mu_{x^2} + t\delta_{x^2}) - (\mu_x + t\delta_x)^2.$$

Then $\beta_{\mathcal{P}} = \beta(0)$, $\beta_{\mathcal{A}} = \beta(1)$, and the first-order approximation is

$$\beta_{\mathcal{A}} - \beta_{\mathcal{P}} = \beta(1) - \beta(0) \approx \beta'(0).$$

Now

$$C'(0) = \delta_{xy} - \mu_y\delta_x - \mu_x\delta_y$$

and

$$V'(0) = \delta_{x^2} - 2\mu_x\delta_x.$$

Since $\beta(t) = C(t)/V(t)$, the quotient rule gives

$$\begin{aligned}\beta'(0) &= \frac{C'(0)V_{\mathcal{P}} - C_{\mathcal{P}}V'(0)}{V_{\mathcal{P}}^2} \\ &= \frac{C'(0) - \beta_{\mathcal{P}}V'(0)}{V_{\mathcal{P}}},\end{aligned}$$

because $C_{\mathcal{P}} = \beta_{\mathcal{P}}V_{\mathcal{P}}$. Substituting the expressions for $C'(0)$ and $V'(0)$, we obtain

$$\begin{aligned}\beta_{\mathcal{A}} - \beta_{\mathcal{P}} &\approx \frac{(\delta_{xy} - \mu_y\delta_x - \mu_x\delta_y) - \beta_{\mathcal{P}}(\delta_{x^2} - 2\mu_x\delta_x)}{V_{\mathcal{P}}} \\ &= \frac{\delta_{xy} + (2\beta_{\mathcal{P}}\mu_x - \mu_y)\delta_x - \mu_x\delta_y - \beta_{\mathcal{P}}\delta_{x^2}}{V_{\mathcal{P}}}.\end{aligned}$$

This is the desired first-order propagation formula. The approximation error is second order in the moment perturbation vector $\boldsymbol{\delta}$, so long as the path $\boldsymbol{\mu}(t)$ remains in a region where $V(t)$ is bounded away from zero. More explicitly, Taylor's theorem gives

$$\beta_{\mathcal{A}} - \beta_{\mathcal{P}} = D\phi(\boldsymbol{\mu}_{\mathcal{P}})[\boldsymbol{\delta}] + \frac{1}{2}\boldsymbol{\delta}^\top H_\phi(\boldsymbol{\mu}_{\mathcal{P}} + \tau\boldsymbol{\delta})\boldsymbol{\delta}$$

for some $0 < \tau < 1$, provided the line segment between $\boldsymbol{\mu}_{\mathcal{P}}$ and $\boldsymbol{\mu}_{\mathcal{A}}$ remains in the domain $V > 0$. Thus the procedure is a genuine first-order delta method-like approximation to the OLS coefficient functional.

This also clarifies why the approximation may fail under severe analytic restrictions. If the moment defect vector is large, or if the analytic restriction substantially changes $V(t)$, the second-order

remainder need not be small. In such cases, this first-order propagation formula can no longer be expected to recover the actual coefficient gap accurately.

Works Cited

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